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PROTOTYPING NEW SCENARIOS FOR FABRICATING STRUCTURAL COMPONENTS FROM RECLAIMED TIMBER

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Introduction

As building practices shift towards bio-based materials, they must also strive to understand the limitations to the extent to which woody biomass can alleviate the destructive effects of post-industrial material procurement and production. Through the development of *Wood ReFramed*, a pavilion imagined as a 1:1 building segment, this paper investigates novel methods for approaching structural applications with reclaimed timber.

Relatively established concepts of resource flow, such as the Circular Economy (CE) and Biomass Cascading (BC), illustrate strategies for enabling greater material efficiency. They propose ecologies organised across technological and biological spheres to create extended material lifetimes, through multiple applications, and portray modes of cyclical matter flows. However, they remain abstracted from the contexts in which material flows take place, neglecting the imperative role of design, and how this engages with prevailing infrastructures for realising alternatives to our current wasteful paradigm.

This study explores actualising the CE and BC theories, by rehearsing the necessary steps required for developing large-scale structures from reclaimed timber. Seeking alternatives to recycling methods of homogenisation and premature incineration, the project maintains timber in its solid form, tackling the associated challenges and uncovering the potential virtues. Through the design and development of the build project *Wood ReFramed*, the paper investigates the following questions:

- How might we design structural components to offer extended lifetimes for reclaimed timber in new material flows?
- How can novel methods interface across existing design and fabrication contexts to increase traction and uptake across industries?

By intercepting what would likely be a transition of material into fuel, the design process engages directly with alternative modes of material sourcing. As an alternative to traditional project timelines, the process interacts with the temporal nature of reclaimed material availability. This paper presents the viability of incorporating reclaimed material in structural components, explored through design, fabrication, and assembly.



State of the art

Recent efforts to assimilate reclaimed timber in structural applications are limited primarily to research investigations. These can be divided into studies that have prioritised a high level of technical assessment, and studies that explore novel systems. The former has typically sought to integrate reclaimed material into existing engineered wood product technologies, such as glulam and CLT (cross-laminated timber) (Rose *et al.*, 2018; Risse *et al.*, 2019). The latter group has investigated broader aspects of reuse, such as locality, and specific relationships between constrained material stock and morphology (Nordby *et al.*, 2014; Bergsagel and Heisel, 2023; Ruan *et al.*, 2023).

The advantage of operating within existing taxonomies is the opportunity to integrate within current structural typologies. However, the studies mentioned demonstrate low material yields (~20%), because of the demands these systems have on uniform material cross-sections and rejection of *defects*. The latter studies offer new methods and material flows, but, through their novelty, are isolated from existing production and building systems.

This research aims to provoke an affinity between the two approaches, investigating the integration of new material flows in existing systems combined with the broader ecologies necessary for developing structural typologies from reclaimed timber.

A structural typology for reclaimed timber

Investigations seeking synergies between the parameters associated with reclaimed timber and structural components has found potential in the patent developed by Hilding Brosenius (Browne *et al.*, 2022). Brosenius's original 'HB Beam' was manufactured in microfactories during the mid-20th century and was applied to a range of building typologies from housing to long-span aircraft hangers (Brosenius, 1990). When applied to a varied stock of reclaimed timber, yields as high as 70% have been reported, arising through reduced processing demands and the ability to include a wider variety of material traits (Browne, 2023).

Structural components in timber comprising multiple parts mechanically fastened together are familiar in traditional truss and frame structures. These were common in a time when materials were costly, and prior to reliable chemical adhesives. However, they still have relevance today, and not only from the perspective of material utilisation.



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Working with waste timber can initiate concerns regarding structural safety, as for those working within the European Union's legislative frameworks it is challenging to assess the structural quality of secondary timber according to its standards. However, it is feasible to design a level of redundancy within a component, where through 'over-structuring' it is possible to utilise materials that would otherwise be discarded. This is akin to strategies employed for global structures to mitigate the possibility for disproportionate collapse. With components containing multiple parts performing individually, a designed progressive brittle failure mode can emulate the ideal ductile failures made possible with steel components and connections. Furthermore, in part-based components, the individual elements can be strategically allocated in response to performance and spatial requirements.



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1. View of the horizontal beams with steel saddles and tension rods to carry the seating area below. © Sara Lohse.

2. Suspending the reclaimed glulam seating onto the tension rods. © Nana Reimers.

3. Inserting the tension rods into the reclaimed glulam seating elements. © Nana Reimers.

4. *Wood ReFramed* on display at the congress. © Sara Lohse.

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As whole components that maintain the elemental identity of their parts, timber species and surface treatments – in all their aesthetic diversity – remain exposed. This offers an opportunity that becomes especially relevant to post-consumer timber, which has been formed through interaction with multiple agencies over time.

Geometry / Topology / Material

The proposed component explores the relationship between geometry, topology, and the availability of material forms. Material, fabrication, and design constraints can be simultaneously evaluated to define optimal component outcomes.

The exterior component geometry is simply described with eight Cartesian coordinates, resembling conventional design space environments and building information norms. Here, rectilinear exteriors are exemplified for their genericness and interchangeability with substitutable components in circumstances where reclaimed material is unavailable. Repetitive geometries pre-empt direct reuse potential, as contemporary examples owe their success to the availability of enough similar components. Within this design space, a determined quantity of material can be allocated based on externalities such as loads, supports, and masked enclaves. The simple two-dimensional topology analysis also defines tension and compression regions. This allocation can be translated into discrete lines, representative of web members, and a boundary representative of upper and lower chords.

The inherited forms of reclaimed material encompass two main variables: the intricate deformations occurring naturally, such as cupping, bowing, and twisting, and the cross-section categories informed by sawmill standards. In the project, these variations are not recorded, described, or modelled. Instead, a domain of cross-sections that can be utilised within the system is defined, significantly reducing the potential complexities arising from part tracking and configuration.

Unrefined parametric relationships

The relationship between discrete lines and varied material leads to an unrefined parametric relationship between material availability and component performance. Web elements are cut with uniform lengths, enabling constant interfacing with the beam system, and avoiding excessive removal of timber. The representation of component topology is simplified, and actual component geometry is only represented in the built form.

As well as allocation, structural assessments using finite element analysis can define optimum material cross-sections for all component parts. While this could enable an especially sensitive relationship between design, material form, and material performance, it would likely also contribute to a reduction in material effectiveness due to increased off-cut. In an unrefined relationship, over-structuring leads to a higher utilisation of input material. Efforts to be more specific about property allocation should work with methods for gathering the structural capacity of material stock. For example, non-destructive test methods could be used to assign grading values to each of the parts, subsequently informing their arrangement within a component.

Joints

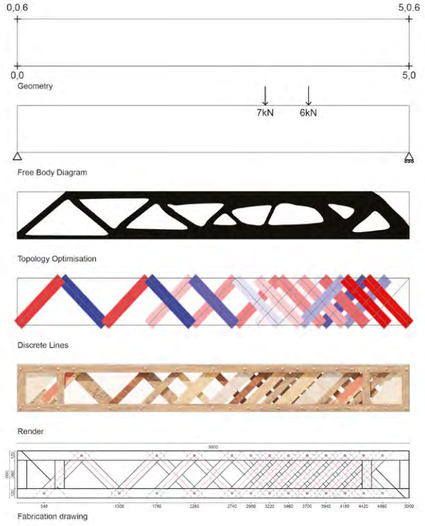
In this project, multi-shear mechanical joints are favoured over adhesives for their ability to connect parts with uneven surfaces. They must, however, be solved intelligently to avoid excessive use of high-impact materials such as steel. Two types of hardwood birch dowels are tested, both with a starting diameter of 19mm, with one set threaded to have an inner diameter of 16mm. Both are effective for multi-shear connections; however, threaded dowels are also capable of locking elements along the dowel's longitudinal axis. Three specimens of each dowel type were tested according to ISO 6891. The unthreaded dowels show a clear advantage in both stiffness and overall capacity, with an elastic limit of 6kN. The unthreaded dowels, however, had an elastic capacity of 4kN.

Wood ReFramed

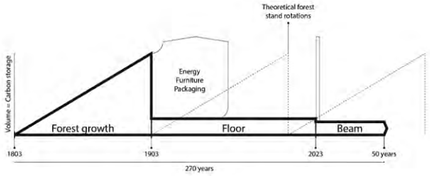
Wood ReFramed was initiated as a 1:1 build project to test the scaling and production of the structural components in collaboration with industry actors. The project is conceived as a building segment, referencing a domain of scales from small buildings to high-rise typologies. The box-like geometry intends to communicate how disordered inventories of reclaimed timber can assimilate with construction conventions. Furthermore, all angles in the project are 45° or 90°, simplifying design and production, as well as offering optimum tension and compression relationships for truss-type structures.

Defining load cases

To test the component typology and methods, realistic building scale loads were a necessary facet for the project. The 15 components are uniform in external geometry, yet each feature a varied distribution and quantity of web elements. This material allocation is mostly informed by structural performance, determined by the specific load cases for each of the frames. To design unique load cases for the repeated frames, a seating area constructed from reclaimed glulam is suspended at 45°. In plan, the centre of gravity lines of the seating steps intersect with the centrelines of the beams, defining the location of suspension brackets. The load case is defined based on an occupancy of 4kN per m² plus the self-weight of the glulam steps, leading to ten point-loads ranging from 1 to 9kN. The maximum force in the web members is calculated to be 4.5kN, well within the elastic range of the regular dowels, but on the upper limit of the threaded dowels. Therefore, regular dowels were applied to shear connections, and threaded dowels are located intermittently to secure assembly.



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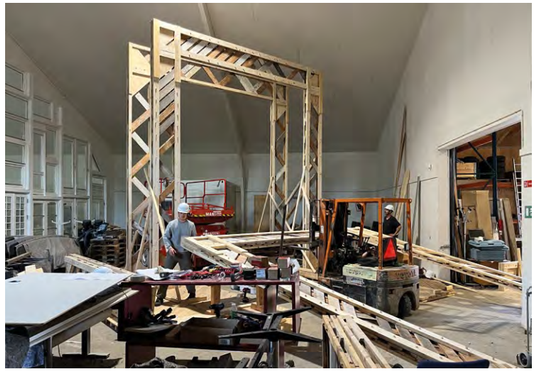
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5. Occupying a rectilinear boundary with different modes of representation.

6. Estimated material timeline spanning 270 years.



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7. Beam during fabrication: assembled component before being released from the jig.

8. Production space during test assembly.

Material search criteria

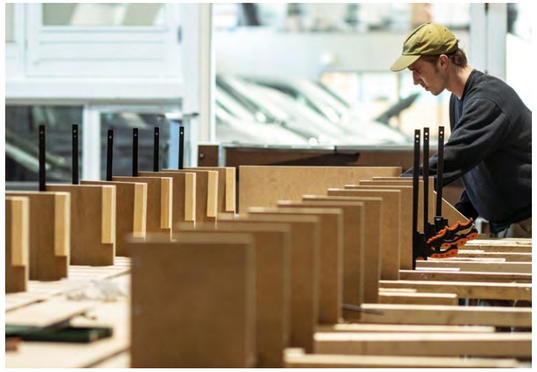
As the project depends on an uncommon material source, a search criterion was developed, informed by the architectural, structural, and detailed design of *Wood ReFramed*.

Danish organisations such as GreenDozer and Genbyg are developing systematic waste collection and sorting strategies intended for use other than recycling. Furthermore, companies such as a:gain are designing new supply chains and stakeholder relations to create building products from established waste streams. However, economic, legislative, and practical constraints remain, leading to issues in valuing materials with high potential. Demolition contractors are critically positioned with access to large quantities of post-consumer material. However, the temporal nature of demolition projects requires them to make quick estimates of a material's value, before deciding whether to invest in careful dismantling, transport, and storage. These speculations traverse a broad market, with many material forms and types passing through their activities.

Jensen Genbrug, the demolition contractor collaborating on this project, at present sells 90% of salvaged materials to private customers and is developing its business to be more attractive to industry. Together with them, a search criterion defining basic material requirements for the project was developed. These key parameters were sent in visual form to all demolition sites in the Copenhagen area for workers to quickly identify compatibility. Minimum lengths, an ideal domain of cross-section, as well as contaminants to be avoided, such as foam, were described. The resulting stock received no formalised structural verification. However, a visual assessment based on the INSTA Nordic standard for timber grading was used to confidently assume design values based on a minimum of C14. Future studies that investigate an enhanced strategic relationship between mechanical properties and component performance could enable more specificity while maintaining the advantages of semi-discrete parts.

In a future where prototypical fringe practices become implemented in everyday projects, modes of secondary material sourcing must operate across significantly longer and more predictable timelines than exemplified here. Current research into the availability of wood within existing structures can quantify usable quantities that will become available post demolition (Kaasalainen *et al.*, 2023). These data sets offer a broader understanding of timber availability, which includes the material already in use. In the same way that the future availability of lumber is today calculated based on forest stand rotations, a holistic assessment can include regions of existing timber buildings where their construction types and estimated lifespan can be included in an overall assessment of timber available for construction.

Adding regional sensitivity to the timber quantities and forms for future construction hints at (new) vernacular architectures. Anthropogenic resources may become synonymous with the terrestrial material availability that has informed the building culture around traditional settlements. The new industries that emerge as a result of valuing post-consumer material will need to be supported by legislative standards that can operate across domains of material parameters, rather than uniform, repetitive requirements.



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Fabrication, assembly, and inhabitation

Many projects employing digital design to develop specific relationships between materials and performance end up with large quantities of unique parts. In this project, adhering to the semi-discrete system of web elements mitigated challenges around traceability, as the variation present in the web elements does not affect their distribution or organisation.

A single jig resembling the precise exterior geometry was used for producing all 15 components. The various parts included in the beam geometrically deviate from the way they are understood in modelling. Slight *bowing* was particularly present in many of the chords, which would traditionally be processed out of individual boards. In this case, during assembly, the beam parts were clamped to the jig as a constant reference geometry for all the components. The layout of the component parts could be described in a simple drawing, with web element placement in a limited number of discrete locations.

The waste timber beams proved significantly more manoeuvrable than the secondary glulams as a result of the strategic allocation of material. Even the examples filled with web elements are half as dense as glulams based on outermost dimensions. This *lightness* potentially enables easier building assembly, and a reduction in foundation magnitude. During assembly, components could be identified by comparing their appearance with project drawings. This suggests that in larger-scale projects with multiple component types of the same outer geometry, individuals could be differentiated and assembled accordingly.

Material origins and futures

As the normative LCA practices draw a system boundary occupying a period from resource extraction to the end of a building's lifetime, they offer a limited representation of material origins. These practices also neglect the former life of secondary materials, and the many interactions that create material conditions are withheld from the assessment of a material's value.

As an investigation into material flow, the notion of components as static objects must be challenged. Uncovering the lines of origin that have defined their emergence are key for understanding the broader ecology of building and depicting the larger terrestrial and cultural consequences of making architecture.

Based on anecdotal evidence from the demolition contractor, we can estimate the material's provenance and geographic movement since its emergence. Depicting the duration of time a material has occupied, we learn that 'new' components are merely convergences of existing matter, convergences informed by a negotiation between the traits of latent material, design, function, and performance. Fig. 6 describes the possible flow of matter as it transitions from emergence to various functions. The y-axis represents material volume, a key metric for assessing effective utilisation and anthropogenic carbon storage. Here, we also see the significant material losses typical during forest harvesting, as only 20% of material is applied to lumber production. This study succeeds in utilising an estimated 75% during the production of web elements.

9. Preparing fabrication jig for the production of all frame components. © Thomas Sinding.

10. Close-up of secondary timber boards. © Xan Browne.

The total distance from demolition site to production site, to exhibition site was less than 100km, demonstrating a significantly shorter supply chain than if virgin materials had been used. As material lifetimes map against forest regeneration, we can start to assess the renewability of material paradigms, as opposed to defining materials as renewable in themselves.

Where material flow is concerned, the components are neither a beginning nor an end. However, determining material availability for future scenarios relies on speculation. Fig. 6 references the 50-year lifespan that is given in the Danish (LCA) standard. Material longevity is integrated into the component's design and is a key aspect for effective cascading. Although the components are designed with a very specific function in mind, the web elements retain the ability to be reorganised, removed, or added to. Reconfigurations may respond to changing functional demands, repair, or disassembly. As timber-only structures, the concentration of material type is homogeneous, and reversible timber joints allow for the separation of wood species. Situating materials in timeframes that are longer than typical building lifetimes raises the question of how we might redesign and reclaim in new methods for creating architecture for the future.

Conclusion

This research investigates potential avenues for reclaimed timber's role in load-bearing structures at the building scale. Exploring new component typologies that are situated within the temporal nature of timber's existence aims to offer insight into the many durations the material can occupy. Determining these durations is a fragile process, as we realise the many sectors that utilise wood, and the need for sufficient material regeneration periods to enable renewable practices. Working across simple Cartesian geometry, performance-informed topology, and inherited material forms intends to enable new matter flows to assimilate into existing practices. These demonstrate that novel methods can integrate with current procedures and support the stakeholders who are positioned to contribute with additional effective resource scenarios.

Optimised buildings are not only about the material quantities they contain, but are assessed in conjunction with a wider ecology of material, production, location, application, and future. Maximising the amount of timber in construction requires a combination of utilising material that is currently undervalued, and designing structures that enable materials to have a life beyond the building, whether that be through long-term component life, or through multiple component lifetimes. The methods presented here intend to correspond holistically with that dialogue, offering insights for better timber practices in the future.

Acknowledgements

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